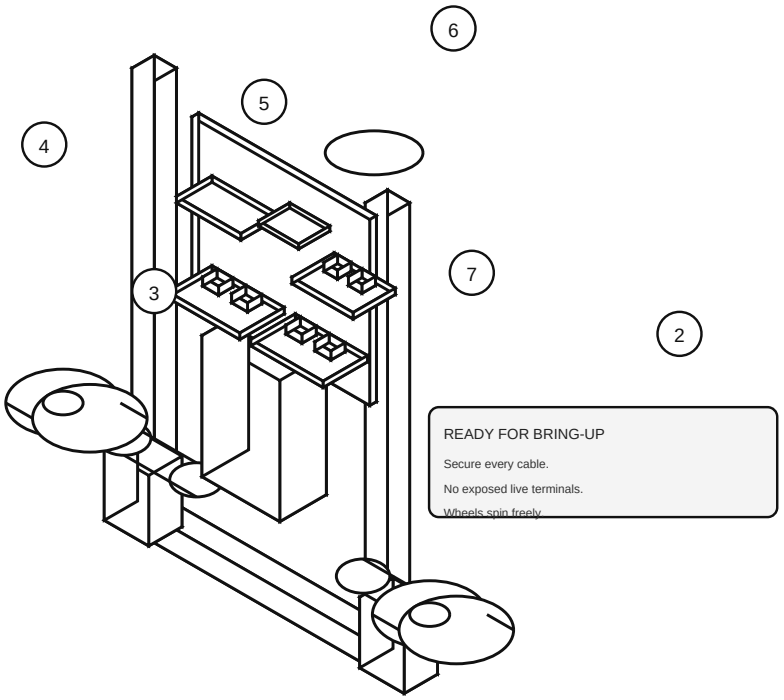
PAULUS v0.1

Two-Wheel Self-Balancing Robot

Illustrated Assembly, Wiring, Bring-Up and Control Manual



Prototype manual • black-and-white vector figures • suggested starting geometry, not a production certification

Scope and safety

What this manual builds

Paulus v0.1 is the physical-control base: an ESP32-controlled two-wheel inverted pendulum using JGB37 encoder gearmotors, BNO085 inertial sensing, two BTS7960/IBT-2 motor drivers, and a 3S battery. The target milestone is stable balancing, closed-loop velocity control, differential steering, and a hardware motor-power emergency stop.

Safety architecture

- The emergency-stop circuit must remove motor power without depending on ESP32 software.
- Place the 15 A fuse close to battery positive. Use the XT60 as the master disconnect.
- Keep hands, loose clothing, pets and fragile objects out of the wheel path during tuning.
- Always begin motor testing with wheels clear of the floor and low PWM duty.
- A sign error in the balance loop creates positive feedback. Verify correction direction while physically restraining the robot.
- Treat the 3S pack as 12.6 V when fully charged; avoid shorts and exposed terminals.

Design rule

Never debug mechanics, power electronics, sensors and control theory simultaneously. Make each layer boring before adding the next one.

Parts and tools

Robot hardware

2x	JGB37-520 12 V encoder gearmotor, target 56:1 (~178 rpm no-load)
2x	Matching wheels/hubs and motor brackets
2x	BTS7960 / IBT-2 motor driver
1x	ESP32 development board (already owned)
1x	BNO085 IMU breakout
1x	3S2P ~11.1 V battery pack with BMS and charger
1x	XL4016 buck converter
1x	12 V automotive relay + 1N4007 flyback diode
1x	22 mm locking normally-closed emergency-stop button
1x	15 A inline blade fuse holder
1x	XT60 connector pair
3x	500 mm 2020 aluminum extrusion
1x	30 x 30 cm, 3 mm acrylic sheet
1 set	2020 angle brackets, M5 T-nuts and screws
1 set	16 AWG silicone wire, Dupont leads, heat-shrink, Faston terminals, straps

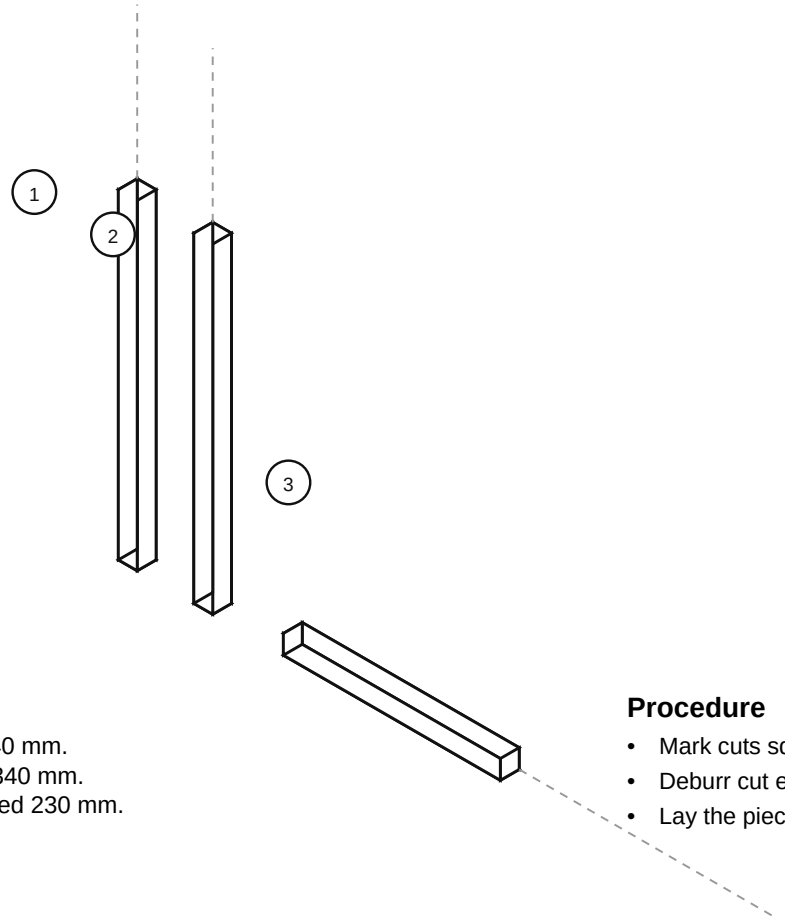
Tools

- Multimeter
- Soldering iron + solder
- Wire stripper and crimp tool
- Drill + suitable acrylic/metal bits
- Hex keys / screwdrivers
- Small M3 hardware and standoffs
- Ruler/caliper and square
- Laptop with ESP32 toolchain
- Optional: bench supply and oscilloscope

Geometry used in these drawings

Suggested starting frame: two 340 mm vertical rails and one 230 mm lower crossmember, cut from the three 500 mm profiles. Battery and deck positions are deliberately adjustable.

Step 1 - Lay out and cut the chassis



Suggested cut lengths

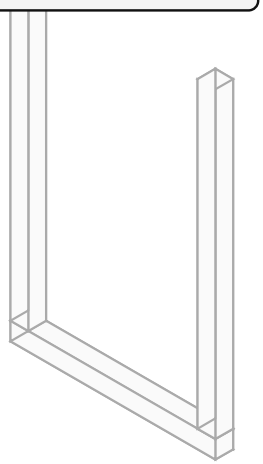
340 mm / 340 mm / 230 mm

REFERENCE

2020 profile: 20 x 20 mm

Original stock: 3 x 500 mm

Keep offcuts for brackets/tests



Callouts

1. Left 2020 vertical rail, suggested 340 mm.
2. Right 2020 vertical rail, suggested 340 mm.
3. Lower 2020 crossmember, suggested 230 mm.

Procedure

- Mark cuts square to the profile. Preserve the offcuts for later brackets or test fixtures.
- Deburr cut ends and clean chips from the extrusion slots.
- Lay the pieces flat in the intended inverted-U arrangement before adding hardware.

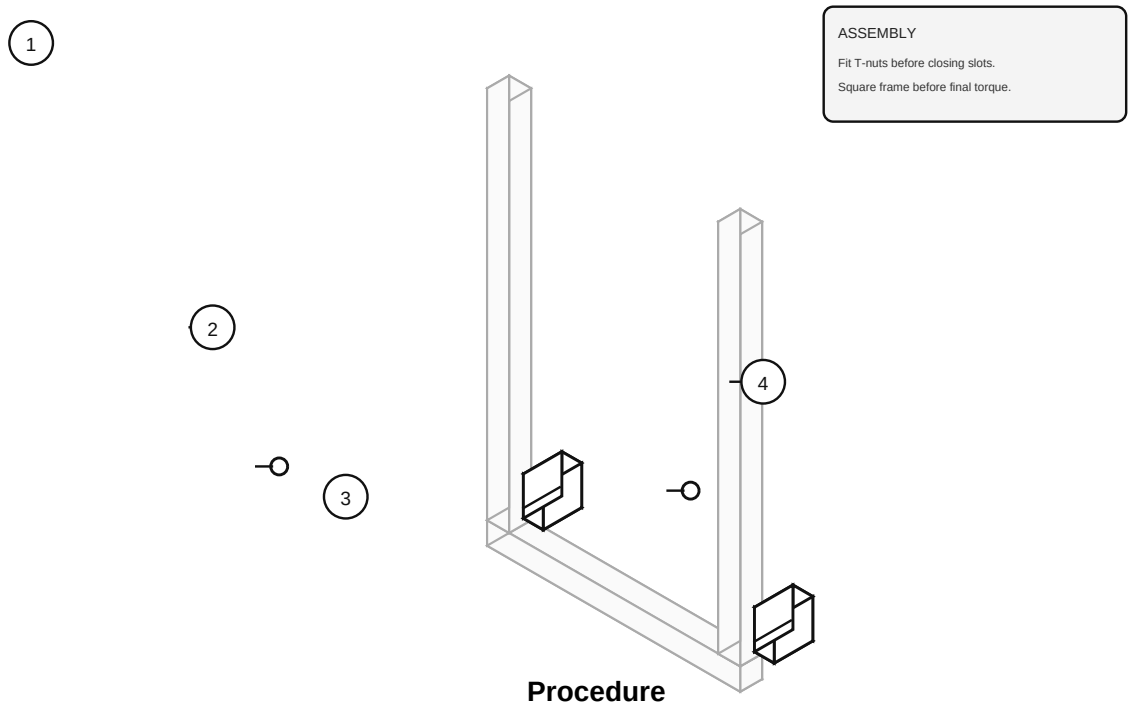
PASS CHECK

Two vertical pieces match in length; crossmember is square; no burrs prevent brackets from seating.

CAUTION

Cut lengths are suggested starting dimensions. If your wheel/bracket geometry demands a different width, change the crossmember before drilling acrylic.

Step 2 - Join the 2020 frame



Callouts

1. 2020 frame pieces.
2. 90-degree corner brackets.
3. M5 T-slot nuts.
4. M5 bracket screws.

Procedure

- Slide the required T-nuts into the slots before closing each corner.
- Fit brackets loosely at both lower corners.
- Use a square or careful diagonal measurement to keep the two vertical rails parallel.
- Only final-tighten after the frame is square.

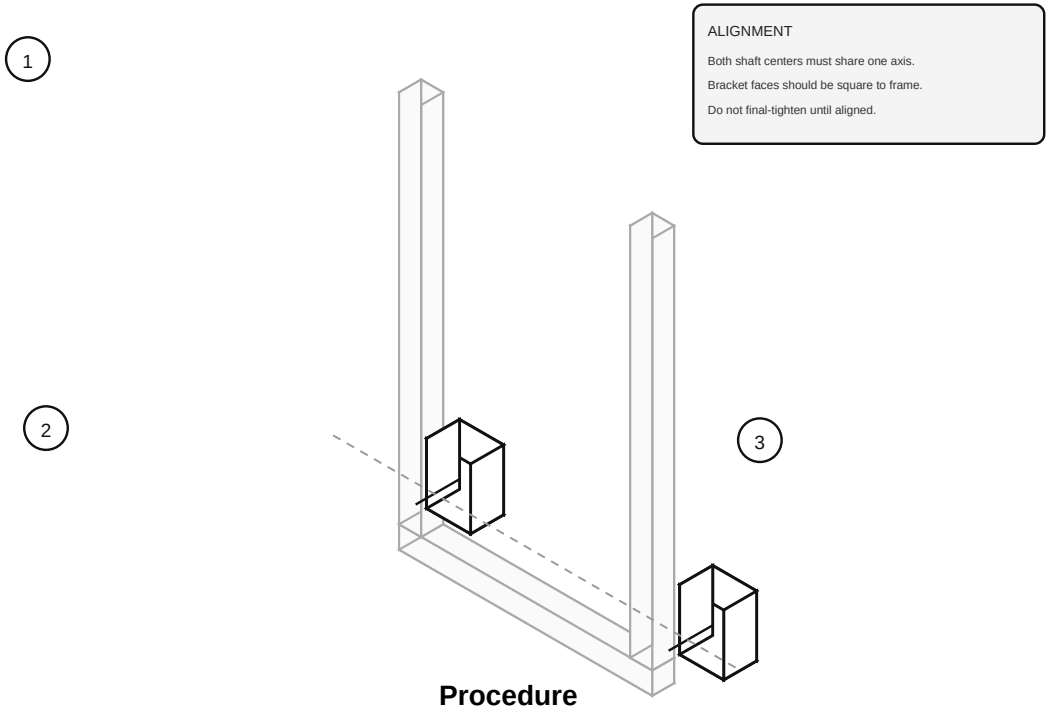
PASS CHECK

Frame sits flat, rails are parallel, and the crossmember does not twist under hand pressure.

CAUTION

Do not over-torque small T-nuts; stripped slot hardware is harder to fix after electronics are mounted.

Step 3 - Install motor brackets



Callouts

1. Lower chassis crossmember.
2. Left motor bracket.
3. Right motor bracket.

Procedure

- Position the two brackets symmetrically at the ends of the lower crossmember.
- Set both shaft centerlines to the same height.
- Check that the bracket faces are square to the frame.
- Leave enough outer clearance for the wheels.

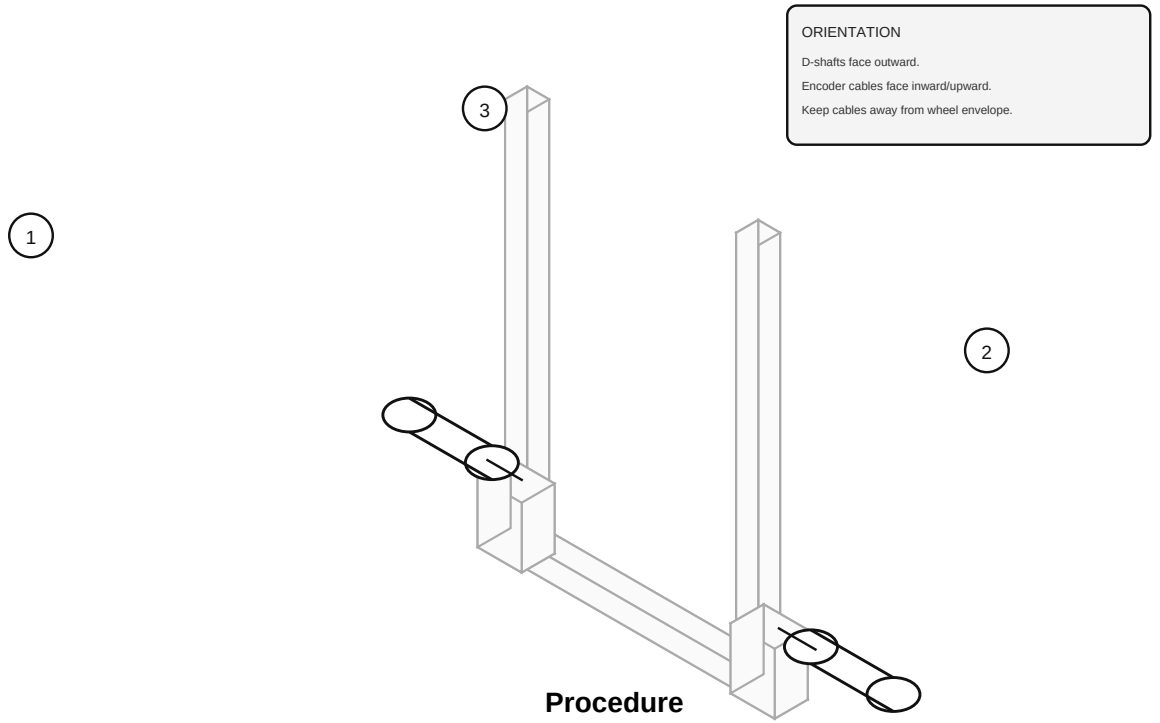
PASS CHECK

A straight reference line through both intended shaft centers crosses both brackets at the same height.

CAUTION

Misaligned wheel axles create constant steering bias and unnecessary bearing loads.

Step 4 - Install both JGB37 motors



Callouts

1. Left JGB37-520 encoder gearmotor.
2. Right JGB37-520 encoder gearmotor.
3. Encoder cable exit / protected routing zone.

Procedure

- Insert each motor into its bracket with the 6 mm D-shaft facing outward.
- Orient encoder wiring inward or upward, never toward the wheel.
- Install and tighten the motor mounting screws evenly.
- Confirm both motors are the same gearbox ratio before proceeding.

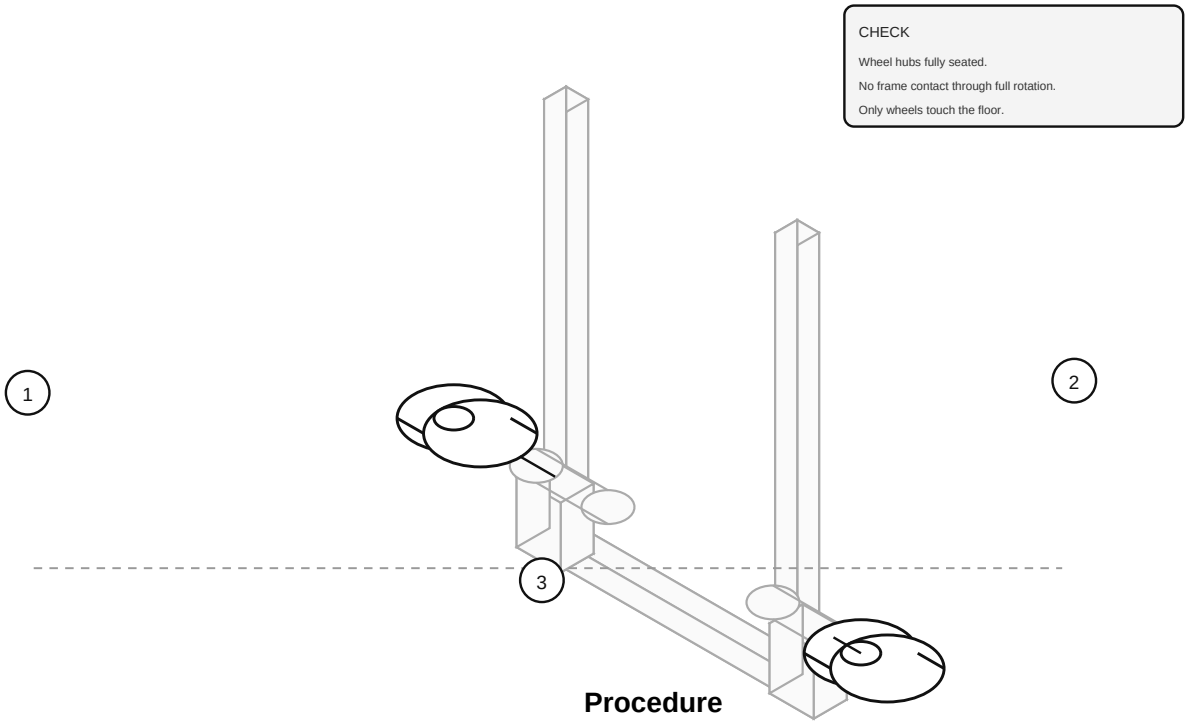
PASS CHECK

Both shafts are horizontal, secure and rotate without the motor body shifting in its bracket.

CAUTION

Do not pull or bend the encoder cable sharply at its exit from the motor.

Step 5 - Install wheels and hubs



Callouts

1. Left wheel/hub.
2. Right wheel/hub.
3. Ground plane / axle reference.

Procedure

- Align each wheel hub with the flat on the 6 mm D-shaft.
- Seat the hub fully without forcing it against the gearbox housing.
- Tighten retaining/set screws securely.
- Spin both wheels by hand and inspect full clearance.

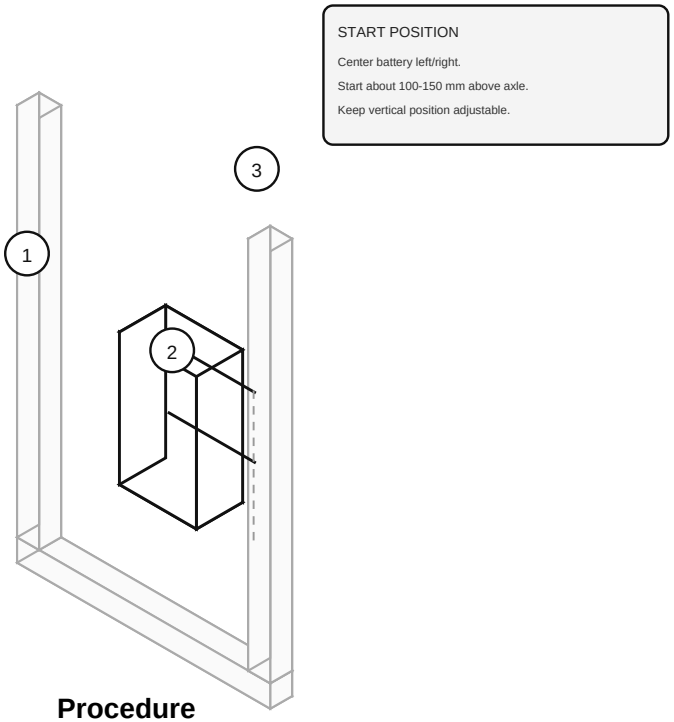
PASS CHECK

Both wheels rotate freely with no frame contact, wobble or significant axial play.

CAUTION

Do not power the motors until wheel clearance has been checked by hand.

Step 6 - Mount the battery provisionally



Callouts

1. 3S2P battery pack.
2. Two battery straps.
3. Vertical adjustment range.

Procedure

- Center the battery left-to-right between the vertical rails.
- Start with its mass roughly 100-150 mm above the wheel axle.
- Use two straps so the pack cannot rotate or slide.
- Keep the mount vertically adjustable for later center-of-mass tuning.

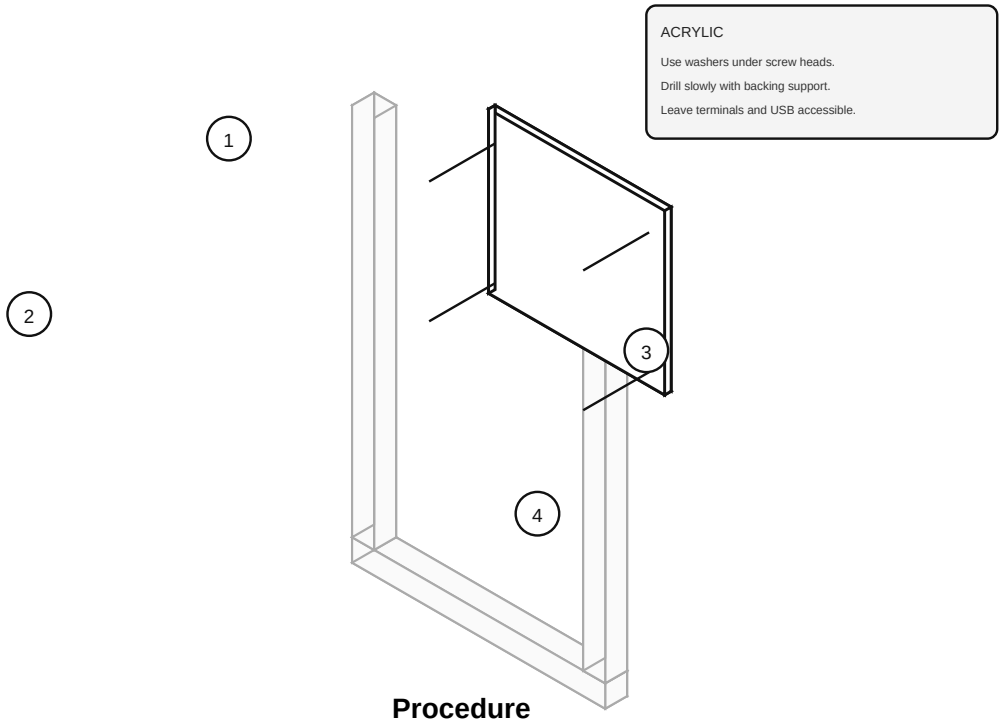
PASS CHECK

Battery cannot move when the frame is shaken gently, but can still be repositioned deliberately.

CAUTION

Do not crush or puncture the pack with brackets, screws or sharp acrylic edges.

Step 7 - Install the acrylic electronics deck



Callouts

1. 3 mm acrylic electronics plate.
2. Standoffs / spacers.
3. Washers and mounting screws.
4. 2020 attachment points.

Procedure

- Cut a practical electronics plate from the 30 x 30 cm sheet; one large plate is not required.
- Mark four mounting points that align with the frame or small support brackets.
- Drill with the acrylic fully supported and low feed pressure.
- Use washers and standoffs; tighten only enough to remove play.

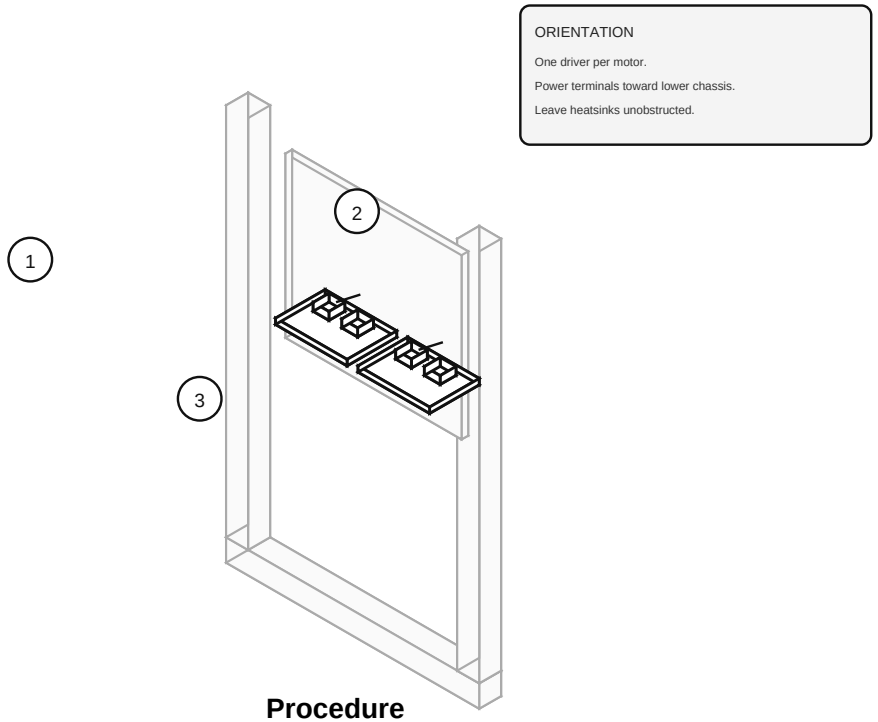
PASS CHECK

Deck is rigid enough for the IMU and does not flex noticeably when connectors are inserted.

CAUTION

Acrylic cracks from point loads. Do not overtighten or drill too close to an edge.

Step 8 - Mount both BTS7960 drivers



Callouts

1. Left BTS7960 / IBT-2.
2. Right BTS7960 / IBT-2.
3. M3 standoffs / cooling clearance.

Procedure

- Mount one driver per motor on standoffs.
- Orient B+/B-/M+/M- screw terminals toward the lower chassis.
- Orient logic headers toward the ESP32 area.
- Leave air around the heatsinks and room for a screwdriver at every terminal.

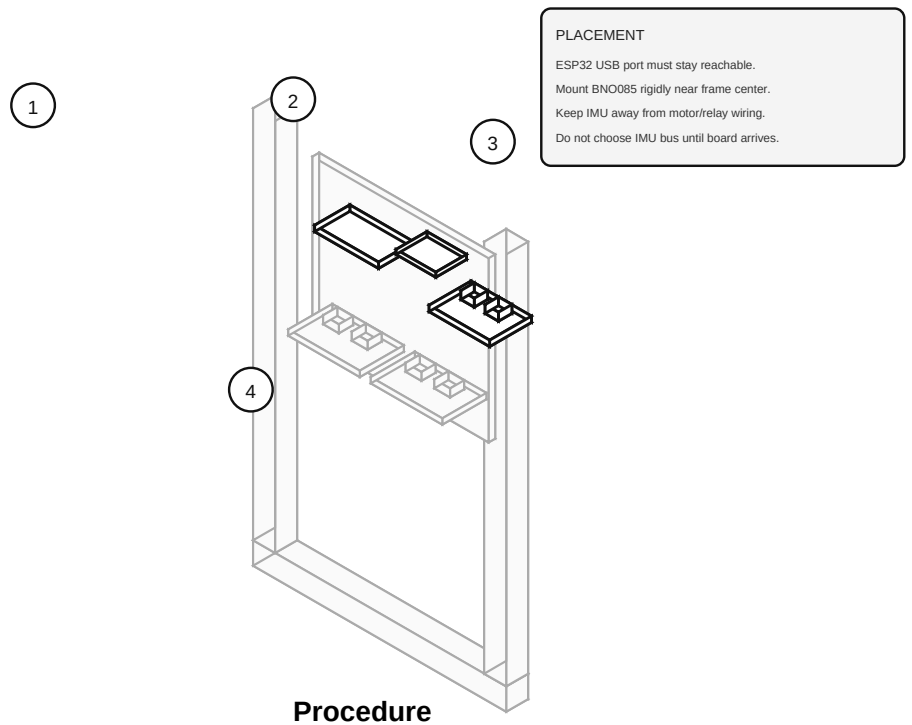
PASS CHECK

Each board is secure, electrically isolated from the frame, and all screw terminals remain reachable.

CAUTION

The advertised 43 A module rating is not a reason to omit the system fuse or proper wiring.

Step 9 - Mount ESP32, BNO085 and XL4016



Callouts

1. ESP32 development board.
2. BNO085 IMU breakout.
3. XL4016 buck converter.
4. Rigid central mounting zone.

Procedure

- Mount the ESP32 so its USB connector remains accessible.
- Mount the BNO085 rigidly near the chassis center and away from motors, relay and heavy current loops.
- Mount the XL4016 where its adjustment control is accessible during bring-up.
- Do not commit to the BNO085 communication bus until the exact breakout markings are inspected.

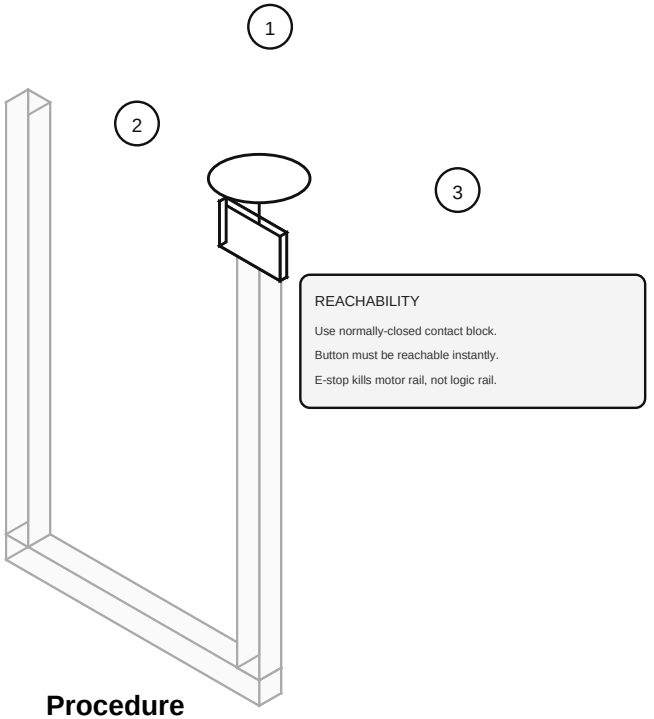
PASS CHECK

All three boards are secure; USB access and regulator adjustment remain possible without disassembly.

CAUTION

BNO08x I2C is known to be troublesome with ESP32-class hosts. Prefer UART or SPI if your breakout exposes those modes.

Step 10 - Install the emergency-stop button



Callouts

1. 22 mm mushroom emergency-stop.
2. Mounting panel / small bracket.
3. Normally-closed contact block.

Procedure

- Choose a position near the upper side of the chassis that is reachable from either side.
- Drill/cut the 22 mm mounting hole or make a small dedicated bracket.
- Insert the switch body, install the retaining nut, and attach the NC contact block.
- Verify the mechanical latch and reset action before wiring.

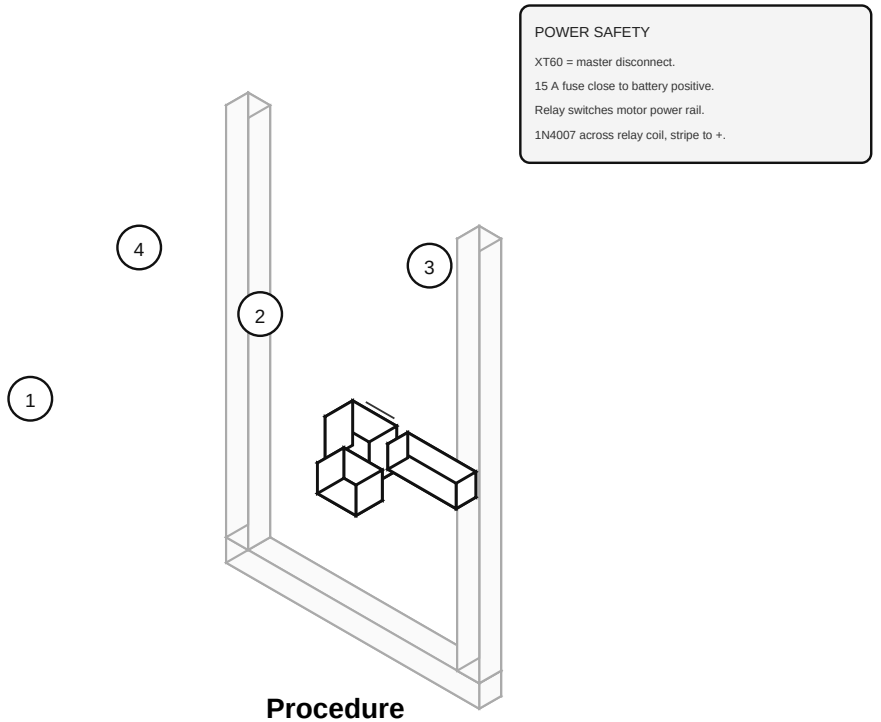
PASS CHECK

Button latches positively; the NC contact opens when pressed and closes only after reset.

CAUTION

This switch is for the relay-coil circuit. Do not route motor current directly through the small E-stop contact block.

Step 11 - Install XT60, fuse, relay and diode



Callouts

1. XT60 master disconnect.
2. 15 A inline blade fuse.
3. 12 V automotive relay.
4. 1N4007 flyback diode across relay coil.

Procedure

- Place the fuse holder physically close to battery positive.
- Mount or secure the relay so its Faston connections cannot touch the aluminum frame.
- Fit insulated 6.3 mm Faston terminals to the relay blades.
- Wire the 1N4007 across the relay coil, striped cathode toward coil positive.

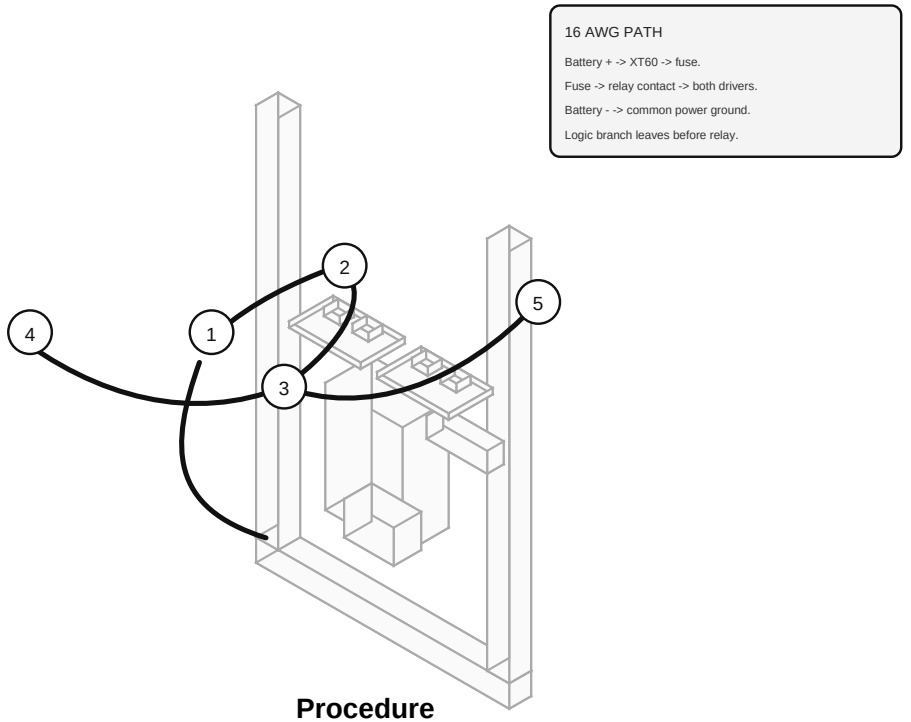
PASS CHECK

Relay is mechanically secure; fuse can be replaced without dismantling the robot; no live Faston blade is exposed.

CAUTION

Confirm the pin diagram printed on your specific relay before using conventional 30/87/85/86 numbering.

Step 12 - Wire the high-current power path



Callouts

1. Battery positive / XT60.
2. 15 A fuse.
3. Relay power contact.
4. Left BTS7960 power input.
5. Right BTS7960 power input.

Procedure

- Use 16 AWG wire for the battery, fuse, relay-contact and motor-driver power path.
- Route battery positive through XT60 -> fuse -> relay contact -> both driver B+ inputs.
- Route battery negative to a compact common power-ground point, then to both driver B- inputs and XL4016 input negative.
- Take the XL4016 input-positive branch upstream of the relay so logic can stay alive when motor power is killed.
- Heat-shrink or insulate every crimp/solder joint and strain-relieve the battery leads.

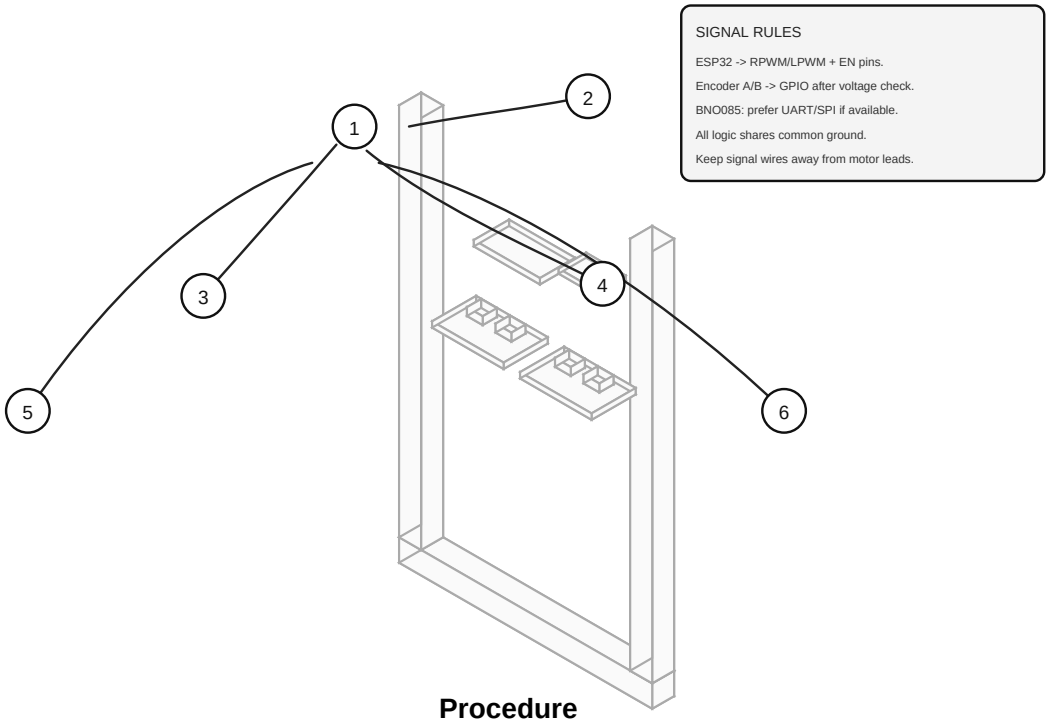
PASS CHECK

With E-stop pressed, driver motor-supply voltage is absent while the logic branch can remain powered.

CAUTION

Do not run motor current through breadboard traces, Dupont jumpers or small PCB headers.

Step 13 - Wire logic, encoders and IMU



Callouts

1. ESP32.
2. BNO085.
3. Left BTS7960 logic header.
4. Right BTS7960 logic header.
5. Left encoder A/B.
6. Right encoder A/B.

Procedure

- Adjust the XL4016 to 5.0 V with a multimeter before connecting the ESP32.
- Connect ESP32 PWM/control signals to RPWM, LPWM, R_EN and L_EN on each driver; share logic ground.
- Measure encoder output voltage before connecting A/B lines directly to ESP32 GPIOs. Add level shifting if needed.
- Choose the BNO085 transport after inspecting the breakout. UART or SPI is preferred if available; I2C may be unreliable with ESP32.
- Physically separate encoder/IMU leads from motor-power wiring and secure every cable.

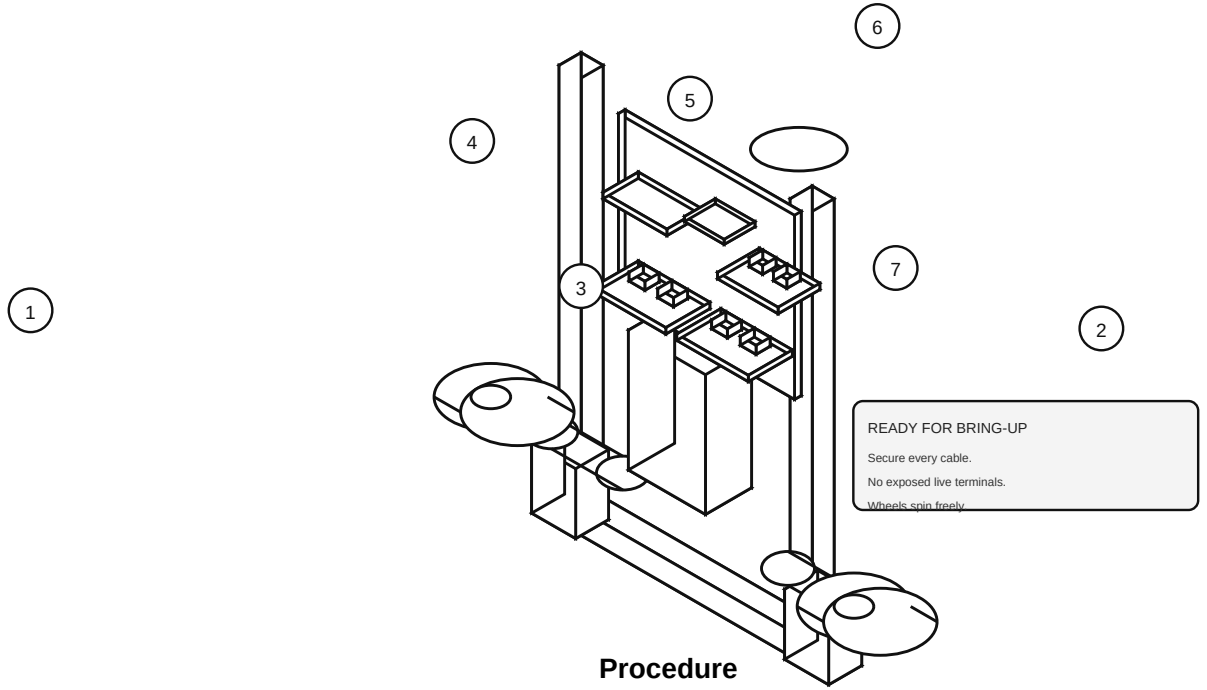
PASS CHECK

Motor-driver logic responds at low PWM, encoders count in both directions, and the IMU streams stable data without bus hangs.

CAUTION

Do not feed 5 V into a 3.3 V-only sensor pin or ESP32 GPIO. Breakout-board regulation/level shifting varies by seller.

Step 14 - Final mechanical assembly and cable management



Callouts

1. Left wheel/motor assembly.
2. Right wheel/motor assembly.
3. Battery.
4. Motor drivers.
5. ESP32 + IMU.
6. Emergency stop.
7. Power-safety hardware.

Procedure

- Tighten structural hardware and re-check wheel alignment.
- Bundle and anchor wiring so nothing can enter either wheel envelope.
- Cover exposed high-current terminals and sharp wire ends.
- Verify battery straps cannot loosen during repeated forward/backward corrections.
- Leave service loops only where needed for connectors; avoid dangling loops.

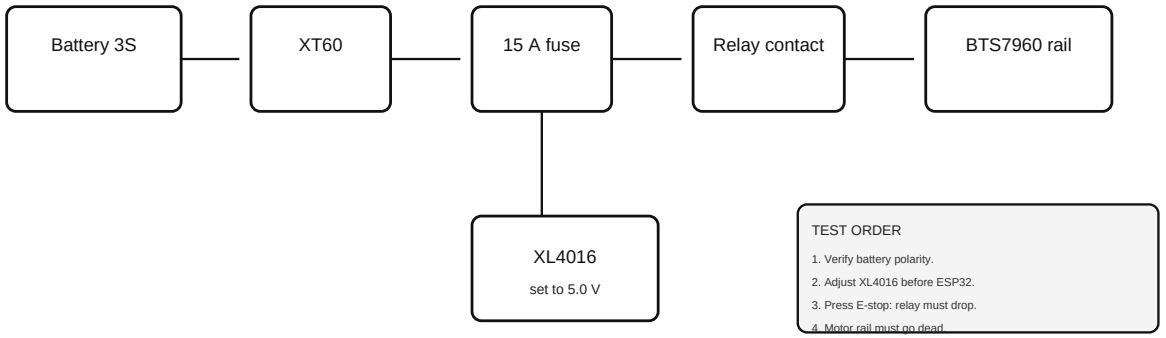
PASS CHECK

Robot can be tilted and shaken gently without any component, battery or cable moving into the wheels.

CAUTION

Do not free-balance around pets or fragile objects. The controller interprets any shove as a disturbance to correct.

Bring-up 1 - Power and emergency-stop verification

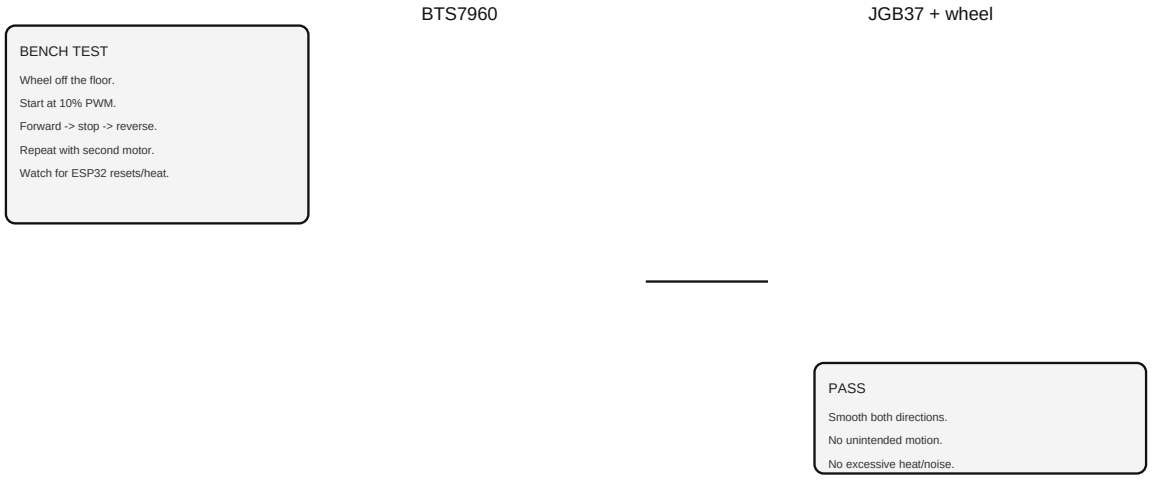


Procedure

- Leave both motor driver outputs disconnected from motors for the first regulator setup.
- Verify battery polarity and measure battery voltage.
- Power only the XL4016 input and adjust its output to 5.0 V. Power down before attaching the ESP32.
- Wire the E-stop in series with the relay coil. Pressing the E-stop must de-energize the relay immediately.
- Measure the motor rail: released E-stop = battery voltage; pressed E-stop = 0 V or disconnected rail.

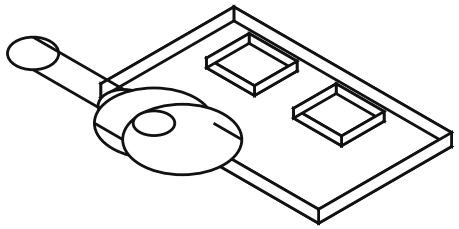
PASS CHECK Pass: the E-stop reliably drops motor power without relying on firmware.

Bring-up 2 - One-motor bench test



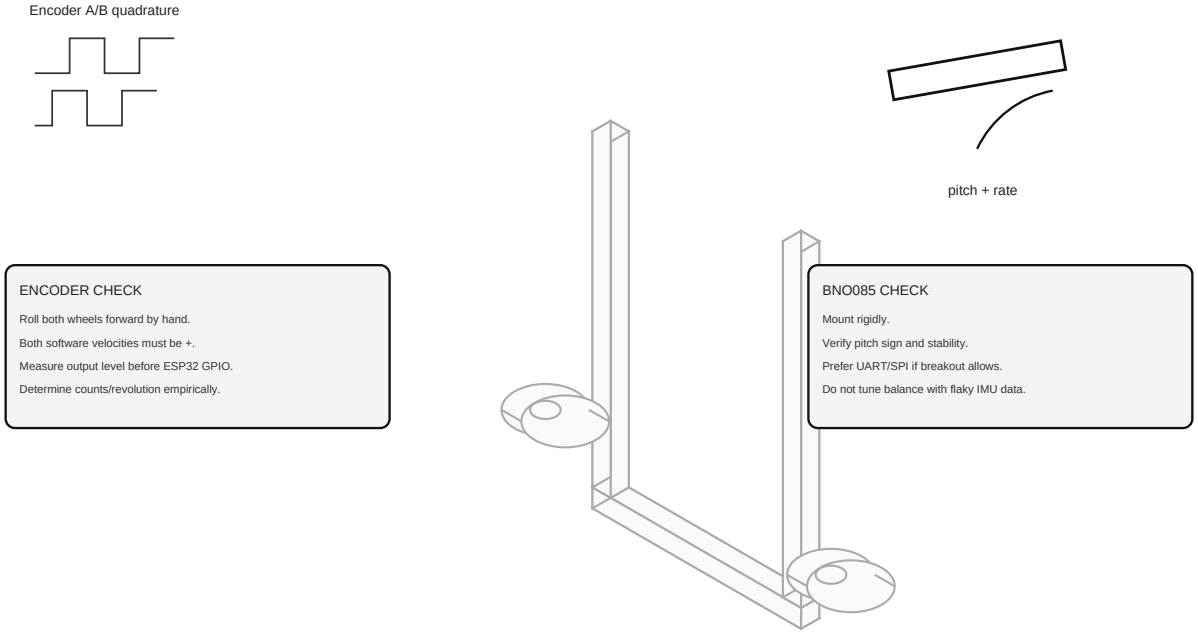
Procedure

- Lift one wheel clear of the table. Connect one motor to one BTS7960.
- Start with low duty (~10%). Command forward -> stop -> reverse -> stop.
- Never drive RPWM and LPWM with nonzero duty at the same time.
- Repeat the test on the second driver/motor pair.
- Watch for unexpected ESP32 resets, driver heating, loose terminal screws or mechanical binding.



PASS CHECK Pass: both motors run smoothly in both directions at low duty and stop on command.

Bring-up 3 - Encoders and BNO085

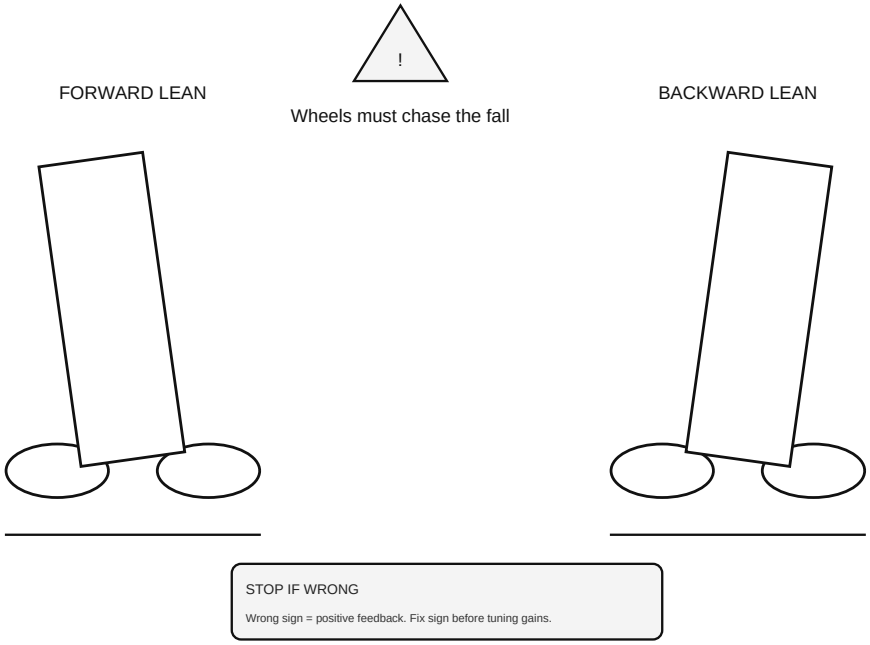


Procedure

- Before ESP32 connection, identify encoder supply, A/B output level and ground. Measure if documentation is unclear.
- Roll each wheel forward by hand and make the software report positive velocity for both sides. Reverse should report negative.
- Determine counts per wheel revolution empirically if the seller does not provide CPR/PPR.
- Bring up the BNO085 separately. Record pitch sign and pitch-rate sign.
- If I2C hangs or drops packets, change transport instead of trying to compensate in the controller.

PASS CHECK Pass: encoder direction conventions are consistent and IMU pitch data is stable while stationary and while hand-tilted.

Bring-up 4 - Feedback sign and first balancing

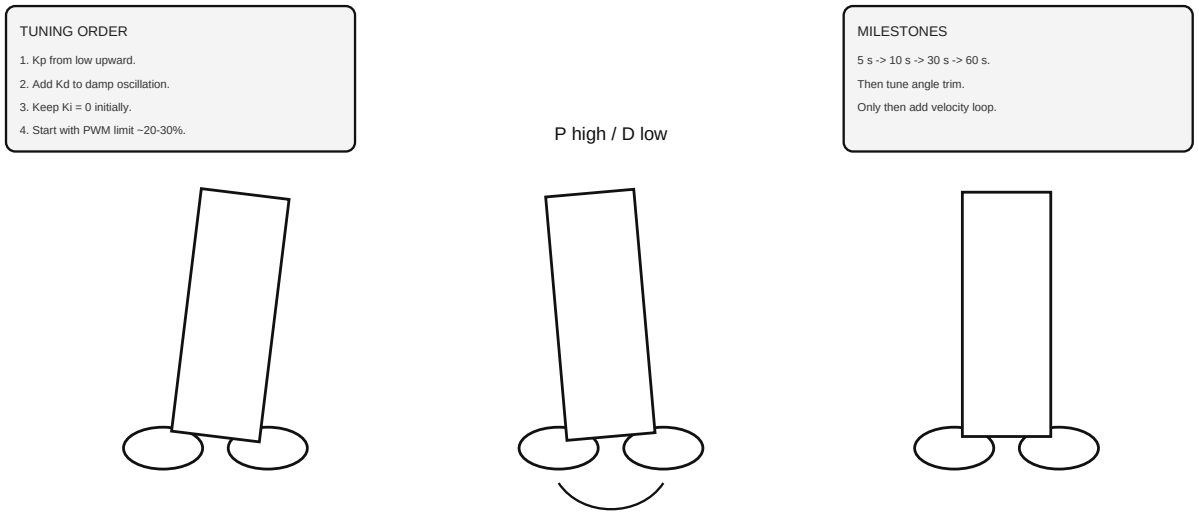


Procedure

- Hold the robot upright with wheels on the floor. Keep one hand ready for the E-stop.
- Lean the chassis slightly forward. The wheels must drive forward underneath the falling center of mass.
- Lean backward. The wheels must drive backward.
- If either correction direction is wrong, fix the sign before tuning any gain.
- Implement a simple inner PD loop first: u = Kp(target_pitch - pitch) - Kd(pitch_rate). Keep Ki = 0 initially.

PASS CHECK Pass: correction direction is always toward the fall, never away from it.

Bring-up 5 - Tune balance, then add velocity and steering



Procedure

- Start with a conservative PWM clamp around 20-30%. Increase Kp from low until the wheels meaningfully chase the center of mass.
- Add Kd to damp oscillation. Too much derivative produces jitter and noise sensitivity.
- Progress through 5 s, 10 s, 30 s and 60 s stationary-balance milestones.
- Find the real neutral angle (angle trim) rather than assuming perfect 0.000 degrees.
- Only after reliable stationary balance, use encoder velocity in an outer loop that changes target pitch.
- Add differential steering around the common balance output: left = u - turn; right = u + turn.

PASS CHECK Pass: Paulus balances for >=60 s, recovers from gentle pushes, drives forward/backward and turns without losing balance.

Troubleshooting matrix

Symptom	Likely cause	First action
Robot instantly accelerates away	Feedback sign reversed	Stop. Reverse motor/controller sign before touching gains.
Robot barely fights the fall	Kp/PWM too low, power issue	Check motor rail and increase authority cautiously.
Violent rapid oscillation	Kp too high / Kd too low / latency	Lower Kp, add damping, inspect loop timing.
Buzzing/jitter near upright	Kd/noise/deadband/backlash	Reduce derivative, inspect filtering and motor deadband.
ESP32 resets when motors start	Power noise / ground bounce	Separate power paths, inspect common ground, add local decoupling.
BNO085 hangs or drops data	Transport issue, especially I2C	Use UART or SPI if your breakout supports it.
Robot creeps while balanced	Neutral angle offset / no velocity loop	Calibrate angle_trim; then add encoder velocity loop.
Always turns one direction	Encoder/motor mismatch or axle alignment	Verify signs, wheel diameter, bracket alignment and velocity gains.
Relay chatters	Weak coil supply / loose E-stop wiring	Measure coil voltage and re-crimp terminals.
Fuse opens repeatedly	Short, stalled motor, wiring fault	Disconnect immediately; inspect motor path before replacing fuse.

Final inspection and v0.1 acceptance

- [ ] Battery secure and undamaged; XT60 disconnect accessible.
- [ ] 15 A fuse installed close to battery positive.
- [ ] Emergency stop mechanically latches and electrically drops motor power.
- [ ] No exposed high-current terminals can touch the aluminum frame.
- [ ] Both wheels rotate freely; motor brackets and axle line remain aligned.
- [ ] All wiring is strain-relieved and outside wheel envelopes.
- [ ] XL4016 verified at 5.0 V before logic connection.
- [ ] Both motor drivers operate forward/reverse independently.
- [ ] Both encoders report correct direction and stable counts.
- [ ] BNO085 data remains stable for extended operation.
- [ ] Robot starts disarmed and refuses to arm when far from upright.
- [ ] Fall detection stops motor commands after a large tilt.
- [ ] Correction sign test passed while restrained.
- [ ] Stationary balance >= 60 seconds.
- [ ] Gentle disturbance recovery works.
- [ ] Closed-loop forward/backward velocity works.
- [ ] Differential steering works without destroying balance.

Technical references used for commissioning notes

- CEVA / Hillcrest BNO08X Datasheet v1.17 - UART-RVC: 115200 baud, 100 Hz report rate.
- Adafruit BNO085 guide - notes BNO08x I2C interoperability problems with ESP32/ESP32-S3; UART/SPI alternatives.
- Espressif ESP32 LEDC documentation - hardware PWM generation for motor-control signals.
- IBT-2/BTS7960 module pinout convention: B+/B-, M+/M-, VCC/GND, RPWM/LPWM, R_EN/L_EN; verify board silkscreen.